

Dr. SONAM MAURYA

Assistant Professor, Department of CSE,
Indian Institute of Information Technology, Pune

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TEACHING EXPERIENCE: 2+ Years

Name of the Organization	Position Held	Duration
Indian Institute of Information Technology (IIIT), Pune, Maharashtra	Assistant Professor (Pay Level-10) in CSE Department	12/12/2022 – Till Date
SRM University AP, Amravati, Andhra Pradesh	Assistant Professor in CSE Department	01/02/2021 – 01/12/2022
Indian Institute of Information Technology (IIIT), Una, Himachal Pradesh	Guest Faculty (on Contract) in School of Computing	16/07/2020 – 21/08/2020
Indian Institute of Information Technology (IIIT), Pune, Maharashtra	Assistant Professor (on Contract) in CSE Department	02/01/2020 – 31/03/2020

RESEARCH INTERESTS

- **Current Research Areas:** Wireless Sensor Networks (WSNs), Internet of Things (IoT), Vehicular Adhoc Networks (VANETs), Delay Tolerant Networks (DTN), Machine Learning, Deep Learning, Image Processing.
- Worked on **International Collaborative Project** entitled “*AMICA- Activity Monitoring Healthcare Android App development*” under the guidance of Mr. **Soumya Kanti Dutta**, CEO and Founder of **Digiotouch Estonia, Europe**, and Research Engineer, EURECOM France.

ACADEMIC BACKGROUND

Degree	Institution	Board/University	CGPA or Percentage	Academic Year
PhD (CSE)	PDPM Indian Institute of Information Technology, Design & Manufacturing, Jabalpur (M.P.)	National Importance Institute established by MoE, Govt. of India	9.7/10 (CGPA)	2015 - 2019
M.Tech (CSE)	Madan Mohan Malviya University of Technology, Gorakhpur (U.P.)	Uttar Pradesh Technical University, Lucknow	79.00 %	2012 - 2014
B.Tech (CSE)	United Institute of Technology, Allahabad (U.P.)	Uttar Pradesh Technical University, Lucknow	78.90 %	2007 - 2011

COURSES TAUGHT

- Data Structure
- Design and Analysis of Algorithms
- Digital Image Processing
- Internet of Things
- Computer Networking
- Mobile and Wireless Networking
- Advanced Wireless Communications

Ph.D. THESIS WORK

Thesis Title: “Energy Efficient Routing Protocols for Heterogeneous Wireless Sensor Networks”.

Thesis Supervisor: Dr. Vinod Kumar Jain (Assistant Professor, CSE Department, IIITDM Jabalpur, M.P.)

Description: In wireless sensor network, sensor nodes are resource-constrained nodes which have limited battery capacity, storage, bandwidth, processing and communication capabilities. These resource limitations impose a challenging task for the researchers to develop energy-efficient low-cost routing protocols with simple operations. Depending upon the specific requirements of different sensor applications, network has either static sink or mobile sink. So, in the thesis, some energy-efficient routing protocols were developed to solve the various routing issues which affect the design and network performance of static sink sensor network and mobile sink sensor network.

PATENTS PUBLISHED / GRANTED

S.N.	Patent Title	Published/ Applied as	Application Number	Patent Status	Published/ Grant Date
1.	Smart Agriculture System using Delay Tolerant Internet of Things	Indian Patent	202241027994	Published	27-05-2022
2.	Smart Wallet with Enhanced Features for Preventing Misuse and Alarm System for the Same	Indian Patent	202241026708	Published	20-05-2022
3.	Board Game Based on Ramayana	Indian Design Patent	350861-001	Prototype submitted for Granting	07-10-2021
4.	Board Game Based on Historical Indian Kings	Indian Design Patent	350860-001	Prototype submitted for Granting	07-10-2021
5.	Board Game Based on King Vikramaditya	Indian Design Patent	350859-001	Prototype submitted for Granting	07-10-2021
6.	Board Game Based on Mahabharatha	Indian Design Patent	350862-001	Prototype submitted for Granting	07-10-2021

RESEARCH PUBLICATIONS

• SCI/SCIE Indexed Journals

- [1] **Sonam Maurya**, Nainil Nandu, Tanay Patel, et al. "A Discrete Cosine Transform based Intelligent Image Steganography Scheme using Quantum Substitution Box", in *Quantum Information Processing* (**Impact factor: 1.965**), *Springer*, 2023. [2nd revised draft submitted]
- [2] **Sonam Maurya**, Sushil Tiwari, et al., "A review on recent developments in cancer detection using Machine Learning and Deep Learning models", in *Biomedical Signal Processing and Control* (**Impact factor: 5.076**), *Elsevier*, Vol. 80, 2023. DOI: <https://doi.org/10.1016/j.bspc.2022.104398>
- [3] Vemula, D.R., Morampudi, M.K., **Maurya, S.** et al., "Enhanced resource provisioning and migrating virtual machines in heterogeneous cloud data center" in *Journal of Ambient Intelligence and Humanized Computing* (**Impact factor: 3.662**), *Springer*, 2022. DOI: <https://doi.org/10.1007/s12652-022-04197-x>
- [4] **Sonam Maurya**, Vinod Kumar Jain, Debanjan Roy Chowdhury, "DA-EERR: Delay Aware Energy Efficient Reliable Routing Protocol for Data Transmission in Heterogeneous Mobile Sink Wireless Sensor Networks" in *Journal of Network and Computer Applications* (**Impact factor: 7.574**), *Elsevier*, Vol. 144, Pages:118-137, ISSN 1084-8045, 2019. DOI: <https://doi.org/10.1016/j.jnca.2019.06.012>.
- [5] **Sonam Maurya**, Vinod Kumar Jain, "Energy Efficient Network Protocol for Precision Agriculture: Using Threshold Sensitive Sensors for Optimal Performance", published in *IEEE Consumer Electronics Magazine* (**Impact factor: 4.135**), Vol. 6, No. 3, Pages: 42-51, 2017. DOI: [10.1109/MCE.2017.2684960](https://doi.org/10.1109/MCE.2017.2684960)
- [6] **Sonam Maurya**, Vinod Kumar Jain, "Fuzzy Based Energy Efficient Sensor Network Protocol for Precision Agriculture", in *Computer and Electronics in Agriculture* (**Impact factor: 6.757**), *Elsevier*, Vol. 130, Pages: 20-37, ISSN 0168-1699, 2016. DOI: [http://dx.doi.org/10.1016/j.compag.2016.09.016](https://doi.org/10.1016/j.compag.2016.09.016).

- **International / National Conferences**

- [1] Shruti Taksali and **Sonam Maurya**, "Scheming of silver nickel magnopsor for Magneto-Plasmonic (MP) activity", in **World Conference on Artificial Intelligence: Advances and Applications (WCAIAA 2023)**, proceeding will be published in **SPRINGER Book Series "Algorithms for Intelligent Systems"**, 2023. [\[Accepted\]](#)
- [2] Vipin Tiwari, Sushil Tiwari and **Sonam Maurya**, "Selection of optimum Grid Size and Unlabeled Data for fingerprinting based Indoor Localization", in **2021- 2nd International Conference on Computational Methods in Science & Technology ICCMST 2021**. pp. 132-136, DOI: [10.1109/ICCMST54943.2021.00037](https://doi.org/10.1109/ICCMST54943.2021.00037)
- [3] **Sonam Maurya**, Vinod Kumar Jain, "Enhanced EECF: An Energy Efficient Coverage Preserving Protocol for Heterogeneous Sensor Networks" in the proceedings of **Conference on Information and Communication Technology (CICT-2018)**, held at PDPM- IITDM Jabalpur, India, during 26-28 October, 2018. DOI:[10.1109/INFOCOMTECH.2018.8722411](https://doi.org/10.1109/INFOCOMTECH.2018.8722411)
- [4] Debanjan Roy Chowdhury, Vinod Kumar Jain, **Sonam Maurya**, "Travel Angle Based Fast Data Dissemination to Relevant Vehicles in VANET" in the proceedings of **41th International Conference on Telecommunications and Signal Processing (TSP-2018)**, held at Athens, Greece during 4-6 July 2018. DOI:[10.1109/TSP.2018.8441394](https://doi.org/10.1109/TSP.2018.8441394)
- [5] **Sonam Maurya**, Vinod Kumar Jain, "EECP: An Energy Efficient Coverage Preserving Protocol for Heterogeneous Sensor Networks" in **2017 IEEE International Conference on Advanced Networks and Telecommunications Systems (IEEE ANTS-2017)** held at Bhubaneshwar, Odisha, India, during 17-20 December, 2017. DOI:[10.1109/ANTS.2017.8384160](https://doi.org/10.1109/ANTS.2017.8384160)
- [6] Sonali Sunil Chaudhary, **Sonam Maurya**, Vinod Kumar Jain, "MAEER: Mobility Aware Energy Efficient Routing Protocol for Internet of Things" in the proceedings of **Conference on Information and Communication Technology (CICT-2017)**, held at Gwalior, India, during 3-5 November, 2017. DOI:[10.1109/INFOCOMTECH.2017.8340624](https://doi.org/10.1109/INFOCOMTECH.2017.8340624)
- [7] Shikha, **Sonam Maurya**, Vinod Kumar Jain, "EHDP: An Efficient Hybrid Data Dissemination Protocol for Dense Irregular Urban Vehicular Ad-hoc Networks" in the proceedings of **Conference on Information and Communication Technology (CICT-2017)**, held at Gwalior, India, during 3-5 November, 2017. DOI:[10.1109/INFOCOMTECH.2017.8340623](https://doi.org/10.1109/INFOCOMTECH.2017.8340623)
- [8] **Sonam Maurya**, Vinayak Gupta, Vinod Kumar Jain, "LBRR: Load Balanced Ring Routing Protocol for Heterogeneous Sensor Networks with Sink Mobility", in **2017 IEEE Wireless Communications and Networking Conference (IEEE WCNC-2017)**, held at San Francisco, CA, USA during 19-22 March, 2017. DOI:[10.1109/WCNC.2017.7925728](https://doi.org/10.1109/WCNC.2017.7925728)
- [9] **Sonam Maurya**, Vinod Kumar Jain, "Threshold Sensitive Region-Based Hybrid Routing Protocol for Precision Agriculture", in **2016 IEEE Wireless Communications and Networking Conference (IEEE WCNC- 2016)**, held at Doha, Qatar, during 3-6 April, 2016, DOI: [10.1109/WCNCW.2016.7552694](https://doi.org/10.1109/WCNCW.2016.7552694)
- [10] **Sonam Maurya**, A.K. Daniel, "RBHR: Region-Based Hybrid Routing Protocol for Wireless Sensor Networks Using AI Technique", in the proceedings of **4th International Conference on Soft Computing for Problem Solving (SocProS-14)**, held at NIT Silchar, India and paper is published in **AISC Series of SPRINGER**, Vol. 336, pp. 37-52, 2015. DOI: [10.1007/978-81-322-2220-0_4](https://doi.org/10.1007/978-81-322-2220-0_4)
- [11] **Sonam Maurya**, A.K. Daniel, "Hybrid Routing Approach for Heterogeneous Wireless Sensor Networks Using Fuzzy Logic Technique", in the proceedings of **4th IEEE Computer Society International Conference on Advanced Computing & Communication Technologies (ACCT-2014)**, held at Rohtak, Haryana, India, during 8-9 February, 2014. DOI: [10.1109/ACCT.2014.81](https://doi.org/10.1109/ACCT.2014.81)

- [12] **Sonam Maurya**, A.K. Daniel, "An Energy Efficient Routing Protocol Under Distance, Energy And Load Parameter For Heterogeneous Wireless Sensor Networks", in the proceedings of **13th International Conference on Information Technology (ICIT-14)**, held at Bhubaneswar, Odisha, India, during 22-24 December, **2014** and paper is published on *IEEE Xplore Digital Library*, ISSN 978-1-4799-8084-0/14, Pages: 161-166, DOI: [10.1109/ICIT.2014.63](https://doi.org/10.1109/ICIT.2014.63)
- [13] **Sonam Maurya**, Anju Verma, A.K. Daniel, "Different Energy Efficient Clustering Based Routing Protocols in Wireless Sensor Networks" published in proceedings of **National Conference on Digital Information & Mobile Communication Technology & its Applications (DIMPCTAP-2014)**, ISSN 978-93-84224-03-5, held at Gorakhpur, India, during 4-5 April, **2014**.
- [14] **Sonam Maurya**, A.K. Daniel, "Energy Efficient Heterogeneous Protocol for Wireless Sensor Network" published in the proceedings of **National Conference on Advances in Systems and Technologies (NCAST-2014)**, held at Gorakhpur, India, during 23-25 January, **2014**.

SOFTWARE SKILLS

- **Operating System:** Windows 11/10, Linux, Ubuntu.
- **Programming Platform:** C, C++, Core Java, MATLAB, Android Studio, Python.
- **Network Simulator:** Omnet++ network simulator, NS-2.
- **Document Preparation Tool:** LaTeX

HARDWARE SKILLS

- Successfully completed 3 weeks FDP workshop (from 16-08-2021 to 04-09-2021) on Internet of Things (IoT) organized by APSSDC, Government of Andhra Pradesh which helped me lot to get practical hands-on experience on **Arduino IDE, Arduino Uno, Lilypad, Raspberry pi development board, Blynk IoT Cloud, Adafruit IoT Cloud** etc. Learnt hardware simulation using "TinkerCAD" to build Voice Assistant based Home Automation System.

ADMINISTRATIVE / PROFESSIONAL RESPONSIBILITIES AND GUIDANCE

- **Member of University International Relations Committee**, SRM University, AP ([OCT 2021 – NOV 2022](#))
- **Department International Relations Committee Member**, SRM University, AP ([FEB 2021 – NOV 2022](#))
- **Department Placement Mentor**, SRM University, AP ([FEB 2021 – NOV 2022](#))
- **Course Coordinator**, Department of CSE, SRM University, AP ([FEB 2021 – June 2021, AUG 2021– DEC 2021](#))
- **Member of Examination Cell**, IIIT Pune ([JAN 2020 – MAR 2020](#))
- **Student Senate Member** of PDPM- IIITDM Jabalpur ([AUG 2017- APRIL 2018](#)).
- **Mess Committee Student Member** of PDPM- IIITDM Jabalpur ([AUG 2017- DEC 2017](#)).
- **Student Member of Hall Executive Committee** of PDPM- IIITDM Jabalpur ([JAN 2015- JULY 2015](#)).
- **Member of Technical Program Committee** for 2021 5th International Conference on Information Systems and Computer Networks (ISCON 2021) held in GLA Mathura during 22nd - 23rd October 2021.
- **Served my duty as a Reviewer** in various reputed Journals:
 - IEEE Transactions on Systems, Man, and Cybernetics: Systems
 - Journal of Network and Computer Applications (Elsevier)
 - KSII Transactions on Internet and Information Systems
 - International Journal of Communication Systems (Wiley Publication)
 - Concurrency and Computation: Practice and Experience (Wiley Publication)
 - IEEE Conference on Information and Communication Technology
 - AEU-International Journal of Electronics and Communications (Elsevier)
 - IEEE Transactions on Systems, Man and Cybernetics: Systems
 - IEEE Access Journal

- **I-Wallet Design Competition Guidance:** I worked along with B.Tech 2nd year SRM-AP Students to design Smart IoT based Wallet solutions and participated in the “I-Wallet Competition” held as a part of TechFest organized by IIT-Bombay. My student Team secured 2nd position *(with a Cumulative 90000 Rs. as a Prize Money)* among over 100 teams participated from various reputed Institutions across the country.

ACHIEVEMENTS / AWARDS

- Received “*Young Scientist Award*” in 11th International Scientist Awards on Engineering, Science and Medicine, held on 17-18 October 2020, Kolkata, India organized by **VDGOOD Professional Association**.
- Received “*Fellowship for Training of Young Scientist*” award by M.P. Council of Science and Technology for my research work in the discipline of Computer Science Engineering and Information Technology, at Rajiv Gandhi Proudhyogiki Vishwavidyalaya, Bhopal during February 28- March 01, 2019.
- Based on academic and PhD research performance, selected to visit **Japan** (from 5-12 December 2018) under “*JENESYS-2018 Science and Technology Exchange*” program (Japan’s friendship ties program advanced by Japanese Government).
- Received 3rd prize for **Poetry Competition** organized under Swachhata Pakhwada-2018, held at PDPM-IIITDM Jabalpur on 2nd October, 2018.
- Received 2nd prize for **Poster Presentation** on the topic “Develop an IoT based solution for Smart Healthcare” in “*Student Design Challenge (SDC)*” event organized during DeW 2017 at PDPM- IIITDM Jabalpur on 11th December, 2017.
- Received 1st prize in **Poster Presentation** on my research topic “*Fuzzy Based Energy Efficient Sensor Network Protocol for Precision Agriculture*”. The poster presentation was organized by IEEE Student Branch PDPM- IIITDM Jabalpur on 3rd October, 2017.
- Awarded “*Certificate of Merit*” in the First year of M.Tech during the academic session 2012-2013.
- Won 1st prize (Mixed team) and 2nd prize (Single event) in **Chess Competition** in the college fest of Madan Mohan Malviya Engineering College, Gorakhpur, during the academic session 2012-2013.
- Received “*Merit Recognition Certificate*” from NIIT for the “*NIIT 6th National IT Aptitude Test (NITAT)*”, held on 21st February, 2010.

WORKSHOP / RALLY PARTICIPATION

- **Attended** two-weeks FDP conducted by SRM University-AP in association with NITTTR Chennai from October 25, 2021 to November 06, 2021 on topics “Collaborative and Active Learning Strategies, Technology Enabled Learning, Outcome based Education, Open Educations Practices, Learner Psychology, Learner engagement and Motivation, Student Assessment and Evaluation, Quality Assurance through Accreditation of HEIs”.
- **Attended** webinar session on “**Who invented this? On the Origin of Disruptive Technologies**” by Dr. Marco Aiello organized as part of the “Distinguished Lecture Series” by Deaprtment of CSE, SRM University-AP on 19th January 2022.
- **Attended** 5 days FDP on “**Quantum Computing and Communications: New Paradigm**” conducted by CSE department, SRM University-AP during 27th December to 31st December 2021.
- **Attended** three-weeks online FDP on “**Internet of Things (IOT)**” conducted by Andhra Pradesh State Skill Development Corporation (APSSDC) during 16th August - 04th September 2021.
- **Participated** in the Department Distinguished Lecture titled “**Neoteric Frontiers in Cloud and Edge Computing**” delivered by Prof. Rajkumar Buyya, Director, Cloud Computing and Distributed Systems (CLOUDS) Lab, The University of Melbourne, Australia and organized by the Department of Computer Science and Engineering, SRM University-AP, India held on 21-July-2021.
- **Attended** one-week online FDP on “**E-Content Development for 21st Century Learners**” conducted by NITTTR Chennai, India during 15-19 June, 2020.
- **Attended and volunteered** in DeW-2017. It was an annual **Design Workshop (DeW)** organized at PDPM IIITDM Jabalpur in collaboration with Ministry of Foreign Affairs, Japan during 09-11 Dec 2017.
- **Participated** in a “**Female Foeticide Awareness Program**” rally initiated by “JAI (Jabalpur Academia Initiative)” on 26th September, 2017.

- **Attended and volunteered** in one-week FDP workshop on “*Network Simulation and Tools (NS-3 and Omnet++)*” conducted by Electronics and ICT Academy (An Initiative of Department of Electronics and ICT, Govt. of India) organized at PDPM-Indian Institute of Information Technology, Design & Manufacturing, Jabalpur, India during 04-09 February, **2017**.
- **Volunteered** in 2 days conference on “*National Conference on Smart Cities (NCSC-2016)*” held at PDPM-IIITDM, Jabalpur, India during 25-26 November, **2016**.
- **Volunteered** in 2 days workshop entitled “*Workshop on Innovation and the Society*” organized by Jabalpur Academia Initiative (6 leading Government Institutions were participated during 5-6 December, **2016**.
- **Attended** one-week FDP workshop on “*Basic Data Mining Algorithms and their Scalability for Big Data*” conducted by Electronics and ICT Academy (An Initiative of Department of Electronics and ICT, Govt. of India) during 16-21 August, **2016**.
- **Attended** one-week FDP workshop on “*Developing Android Applications for Fun & Profit*” conducted by Electronics and ICT Academy (An Initiative of Department of Electronics and ICT, Govt. of India) during 14-20 December, **2015**.
- **Attended** two days workshop on “*Cyber Security and Malware Analysis (Level-1)*” conducted by Indian Servers, Software Development & Web Hosting Company, held on 20-21 March, **2015**.
- **Attended and volunteered** in four days symposium on “*Next Generation Networks*” organized by PDPM-Indian Institute of Information Technology, Design & Manufacturing, Jabalpur, during 12-15 March, **2015**.
- **Attended** four days “*International Conference on Advances in Computing, Communications and Informatics*” held at Galgotia College of Engineering & Technology, Greater Noida, during 24-27 September, **2014**.
- **Attended** four days workshop cum training courses on “*AVR Architecture based Embedded System & Robotics*” conducted by BRiCS, IIT Kanpur, in the year **2009**.

ACADEMIC SOCIAL NETWORKING

1. **ResearchGate:** https://www.researchgate.net/profile/Sonam_Maurya
2. **Google Scholar:** <https://scholar.google.co.in/citations?hl=en&user=aw3QZNEAAAAJ>

PERSONAL INFORMATION

Date of Birth: 04th November, 1990

Father's Name: Narendra Prasad Maurya

Marital Status: Married

Language Known: Hindi, English

Nationality: Indian